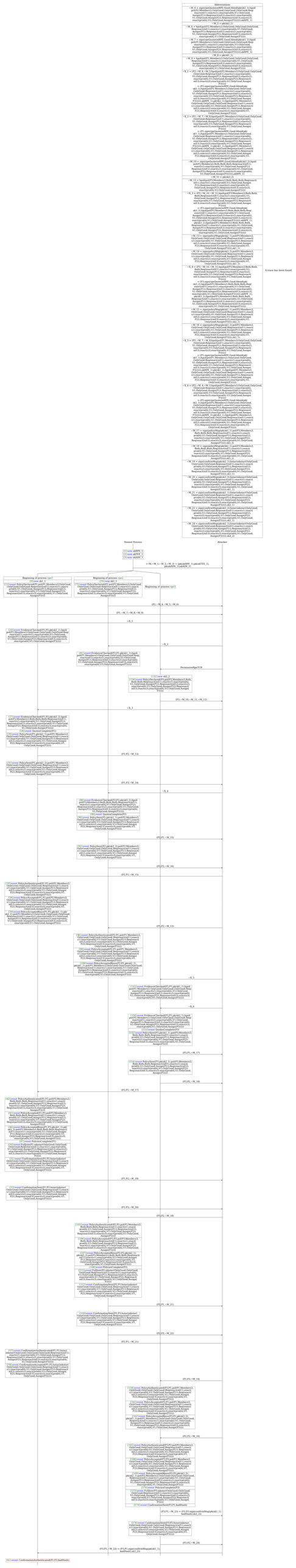

Abbreviations

~M_4 = sign(rpeQuote(mRPE,Good,hbind(pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1)

~M_5 = pk(sk2_1)

~M_6 = hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))

~M_7 = sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1)

~M_8 = pk(sk1_1)

~M_9 = hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))

~X_1 = (P2,~M_4,~M_5,hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
= (P2,sign(rpeQuote(mRPE,Good,hbind(pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~X_2 = (P1,~M_7,~M_8,hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
= (P1,sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~M_10 = sign(rpeQuote(mRPE,Good,hbind(pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1)

~M_11 = pk(sk3_2)

~M_12 = hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))

~X_3 = (P3,~M_10,~M_11,hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
= (P3,sign(rpeQuote(mRPE,Good,hbind(pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~M_13 = sign(policyMsg(pk(sk2_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk1_1)

~M_14 = sign(policyMsg(pk(sk3_2),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk1_1)

~X_4 = (P3,~M_10,~M_11,hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
= (P3,sign(rpeQuote(mRPE,Good,hbind(pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~M_15 = sign(policyMsg(pk(sk1_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk2_1)

~M_16 = sign(policyMsg(pk(sk3_2),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk2_1)

~X_5 = (P1,~M_7,~M_8,hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
= (P1,sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~X_6 = (P2,~M_4,~M_5,hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
= (P2,sign(rpeQuote(mRPE,Good,hbind(pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~M_17 = sign(policyMsg(pk(sk1_1),pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk3_2)

~M_18 = sign(policyMsg(pk(sk2_1),pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk3_2)

~M_19 = sign(confirmMsg(pk(sk2_1),hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))),sk1_1)

~M_20 = sign(confirmMsg(pk(sk3_2),hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))),sk1_1)

~M_21 = sign(confirmMsg(pk(sk1_1),hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))),sk2_1)

~M_22 = sign(confirmMsg(pk(sk3_2),hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))),sk2_1)

~M_24 = sign(confirmMsg(pk(sk2_1),hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))),sk3_2)

A trace has been found.

Honest Process
Attacker
{1}new akRPE_1
{2}new akTEE_1
{3}new akADV_1
(~M,~M_1,~M_2,~M_3) = (pk(akRPE_1),pk(akTEE_1),pk(akADV_1),akADV_1)
Beginning of process rpe1
{9}new sk1_1
{11}event PolicyDeclared(P1,pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
Beginning of process rpe2
{70}new sk2_1
{72}event PolicyDeclared(P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
Beginning of process rpe3
(P2,~M_4,~M_5,~M_6)
(P1,~M_7,~M_8,~M_9)
~X_1
{20}event EvidenceChecked(P1,P2,pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
~X_2
{81}event EvidenceChecked(P2,P1,pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
PermissiveRpeTCB
{132}new sk3_2
{134}event PolicyDeclared(P3,pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P3,~M_10,~M_11,~M_12)
~X_3
{27}event EvidenceChecked(P1,P3,pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{28}event QuotesComplete(P1)
{29}event PolicySent(P1,pk(sk2_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P1,P2,~M_13)
{31}event PolicySent(P1,pk(sk3_2),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P1,P3,~M_14)
~X_4
{88}event EvidenceChecked(P2,P3,pk(sk3_2),hpol(pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{89}event QuotesComplete(P2)
{90}event PolicySent(P2,pk(sk1_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P2,P1,~M_15)
{92}event PolicySent(P2,pk(sk3_2),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P2,P3,~M_16)
(P2,P1,~M_15)
{35}event PolicyAuthenticated(P1,P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{38}event PolicyAccepted(P1,P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{39}event PolicyAcceptedKeys(P1,P2,pk(sk1_1),pk(sk2_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P1,P2,~M_13)
{96}event PolicyAuthenticated(P2,P1,pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{99}event PolicyAccepted(P2,P1,pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{100}event PolicyAcceptedKeys(P2,P1,pk(sk2_1),pk(sk1_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
~X_5
{143}event EvidenceChecked(P3,P1,pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
~X_6
{150}event EvidenceChecked(P3,P2,pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{151}event QuotesComplete(P3)
{152}event PolicySent(P3,pk(sk1_1),pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P3,P1,~M_17)
{154}event PolicySent(P3,pk(sk2_1),pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P3,P2,~M_18)
(P3,P1,~M_17)
{42}event PolicyAuthenticated(P1,P3,pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{45}event PolicyAccepted(P1,P3,pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{46}event PolicyAcceptedKeys(P1,P3,pk(sk1_1),pk(sk3_2),pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{47}event PoliciesComplete(P1)
{50}event FullJoin(P1,mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{51}event ConfirmationSent(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
(P1,P2,~M_19)
{53}event ConfirmationSent(P1,P3,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
(P1,P3,~M_20)
(P3,P2,~M_18)
{103}event PolicyAuthenticated(P2,P3,pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{106}event PolicyAccepted(P2,P3,pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{107}event PolicyAcceptedKeys(P2,P3,pk(sk2_1),pk(sk3_2),pol(P3,Members3,Both,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{108}event PoliciesComplete(P2)
{111}event FullJoin(P2,mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{112}event ConfirmationSent(P2,P1,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
(P2,P1,~M_21)
{114}event ConfirmationSent(P2,P3,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
(P2,P3,~M_22)
(P2,P1,~M_21)
{57}event ConfirmationAuthenticated(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{59}event ConfirmationAccepted(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
(P1,P3,~M_14)
{158}event PolicyAuthenticated(P3,P1,pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{161}event PolicyAccepted(P3,P1,pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{162}event PolicyAcceptedKeys(P3,P1,pk(sk3_2),pk(sk1_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P2,P3,~M_16)
{165}event PolicyAuthenticated(P3,P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{168}event PolicyAccepted(P3,P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{169}event PolicyAcceptedKeys(P3,P2,pk(sk3_2),pk(sk2_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{170}event PoliciesComplete(P3)
{173}event FullJoin(P3,mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{174}event ConfirmationSent(P3,P1,badHash)
(P3,P1,~M_23) = (P3,P1,sign(confirmMsg(pk(sk1_1),badHash),sk3_2))
{176}event ConfirmationSent(P3,P2,hstar(mkstar(OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
(P3,P2,~M_24)
(P3,P1,~M_23) = (P3,P1,sign(confirmMsg(pk(sk1_1),badHash),sk3_2))
{62}event ConfirmationAuthenticated(P1,P3,badHash)